

Session 23 : Research & Build

The actual investigation happens here. Today you move from planning to doing - reading deeply, exploring hands-on, and drafting the core sections of your paper.



Recap : Last Session (S22)

- Picked your domain
- Sharpened your research question
- Found candidate sources
- Committed to a hands-on exploration plan

In homework, you drafted your **Related Work** and **Introduction** sections.

Today is **Research & Build**. The actual investigation happens here. You move from preparation into the real work of your paper.

Today's Two Threads

Your investigation runs on two parallel tracks - both are essential, and neither alone is the investigation.



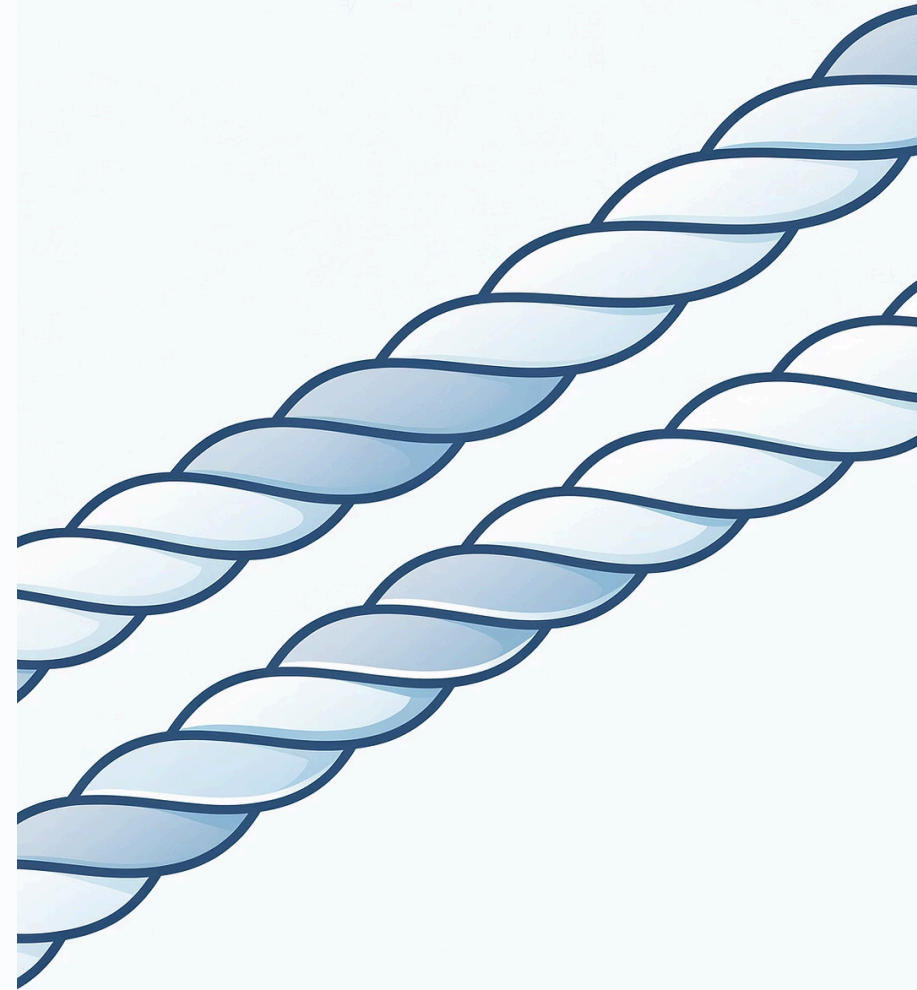
Literature Thread

Read your sources deeply. Pull out specific findings. Build the foundation for your Results section.



Hands-On Thread

Use a tool to gain firsthand experience with how AI works in your domain. Observe what it does, where it struggles, what surprises you.



Today's Goals

By the end of this session, you'll have made real, measurable progress across every major section of your paper.

01

Sources Read Deeply

Specific findings extracted from each source

02

Exploration Completed

Hands-on tool experience documented

03

Methodology Drafted

Your investigation process written up

04

Results Started

Both threads beginning to come together

05

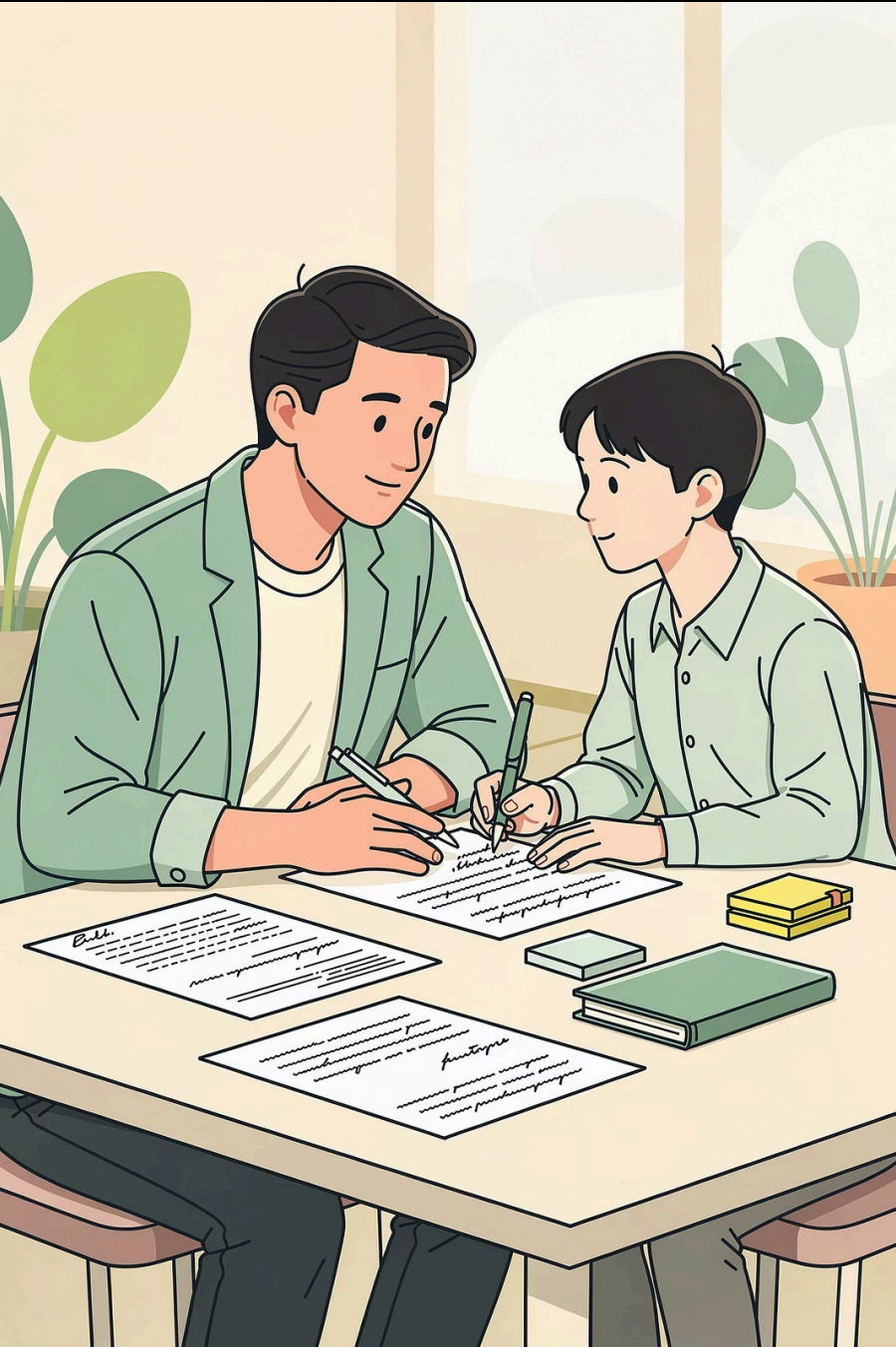
Related Work Refined

Sharpened based on deeper reading

06

User Testing Plan

3-5 testers identified for homework



Homework Review

What Your Mentor Will Read

Your mentor will read your **Related Work** and **Introduction** drafts from S22 homework. Expect specific, targeted feedback.

What They'll Flag

- Sentences that sound generated rather than written by you
- Claims without citations
- Generic phrasing where specific would be stronger

Your Job Right Now

Take notes on every piece of feedback. You'll address these issues during the polish work in **Session 24**.

The Deep Literature Read

In S22 you confirmed your sources existed and were credible. Today you read them properly. For each source, extract the following:



Main Finding

The source's main finding or argument about AI in your domain



Specific Facts

Statistics, examples, or cases you can cite directly in your paper



Direct Quotations

Short - under 15 words each - only when the exact wording matters



Source Relationships

How this source fits with others - does it agree, contradict, or extend?

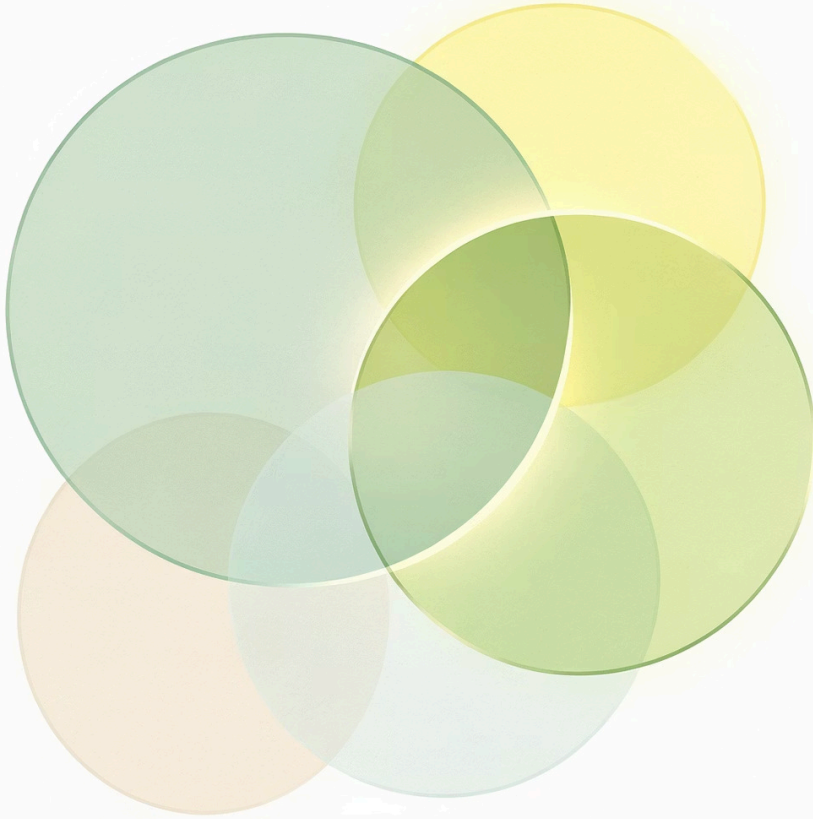
❗ Use **NotebookLM** as your primary tool - it's grounded in source material and less likely to fabricate content than general AI models.

Reading One Source - Workflow

For each of your 5–8 sources, work through these steps in order. Don't skip ahead.

01	02	03
Upload to NotebookLM	Ask for the Main Finding	Ask for Specific Examples
Add the source to your NotebookLM workspace	"What is this source's main finding about AI in [your domain]?"	"What specific examples, statistics, or cases does it provide?"
04	05	
Ask for Limitations	Save to Notes	
"What limitations or concerns does it raise?"	Record answers in a notes document — you'll cite from these later	

⊗ Don't skim. Don't trust a summary. Get into the actual content.



Synthesizing Across Sources

Once you've read each source, look for patterns. This is what synthesis means - your Related Work isn't a list of source summaries, it's a synthesis of what the field collectively knows.

Where Do They Agree?

What do your sources collectively confirm about AI in your domain?

Where Do They Disagree?

What tensions or contradictions exist between sources?

What Does Source A Say That Source B Doesn't?

Unique contributions from individual sources that fill gaps

Repeated Examples?

Where do you see the same finding cited in multiple places?

A Pattern to Watch For

In most domains, sources will fall into rough groupings. Organize your Related Work around these groupings not source by source.

What AI Does

Sources that describe what AI does in this domain

How Well It Works

Sources that report on effectiveness and performance

Concerns

Sources that raise issues about bias, safety, or ethics

Policy & Regulation

Sources that discuss governance and oversight



Organizing around these groupings is what makes Related Work feel like **real research**.

The Hands-On Exploration

Now you do your hands-on exploration

What This Is

A learning activity. Its purpose is to give you firsthand experience with how AI works in your domain.

Your observations become part of your paper.

What the Paper's Contribution Is

Your investigation of how AI is being applied in your domain. The exploration *informs* that investigation.

The exploration is evidence. The investigation is the argument.



Setting Up the Exploration

Before you start, write down what you're about to do. This becomes the basis for your Methodology section.



Which Tool

Name the specific tool you're using



What You'll Try

What you're going to create or attempt



What You Expect

What you predict will happen before you start



What You're Watching

What you're paying attention to during the exploration

 This setup note takes 2-3 minutes. Don't skip it. Documenting before you start is what separates a research exploration from playing around.

While You Explore - Pay Attention

As you do the exploration, actively observe and take notes. Specific observations are more useful than general impressions.

What does the tool do well?

Where does it struggle or fail?

Does it surprise you in any way?

Does it match what your sources said?

Does it contradict anything you read?

 Take notes **as you go** - don't wait until the end to try to remember what happened.

Examples of Exploration in Practice

Here's what hands-on exploration looks like across three different domains. The point isn't to get a perfect result — it's to learn what the technology actually does.

Domain	What the student does	What they pay attention to
Food allergy management	Sets up a Teachable Machine classifier with peanut and non-peanut food images	Where it correctly identifies allergens; where it gets confused; whether confidence matches accuracy
Music education	Uses ElevenLabs to generate the same sentence in five emotional tones	Whether emotional tones are clearly different; what each tone changes about the voice; whether listeners can identify the intended emotion
Wildlife identification	Uses iNaturalist to identify 15 photos of local plants and animals	Which species it identifies confidently; which it gets wrong; how it handles unclear photos

Documenting Your Observations

After the exploration, write down what you observed. Be specific - vague impressions won't help you write Results.



What It Did Well

One or two concrete examples of success



Where It Struggled

One or two concrete examples of failure or limitation



What Surprised You

And specifically why it surprised you



What You Learned

What this taught you about AI in your domain

These observations become part of your Results and Bias & Safety sections later.



Drafting Methodology

Methodology describes how you investigated your topic. You're not justifying engineering choices - you're documenting your investigation process so a reader can understand how you reached your findings.

Literature Investigation

- What sources you read
- How you found them
- How you verified them
- How you synthesized across them

Hands-On Exploration

- What tool you used
- What you set up
- What you tried
- What you paid attention to

Methodology Structure

A useful structure for your Methodology section

Brief Restatement

What you were
investigating

Literature Workflow

Description of your
literature research
process

Exploration Setup

Description of your
hands-on exploration

Synthesis Note

How you combined
findings across both
threads

Methodology Example

Here's how a Methodology section might read for the food allergy management investigation:

"This investigation drew on two threads. For the literature thread, I identified five candidate sources through Perplexity searches on terms related to AI and food allergy management for children. I filtered for credibility using a five-point checklist, then verified content using NotebookLM. From these sources, I extracted findings about applications, effectiveness, and concerns. For the hands-on thread, I set up a small classifier using Teachable Machine, training it to distinguish images containing peanuts from images without peanuts. The exploration was meant to give me firsthand experience with image-based AI in this domain, not to test a hypothesis. I observed where the classifier worked well, where it struggled, and where its behavior matched or contradicted what the literature described."

❗ This is one example, not a template you have to follow. **Use your own voice.**

Drafting Results

Results synthesizes the two threads of your investigation. This is where the literature and the exploration come together.

From Literature

What your sources collectively show about AI in your domain

From Exploration

What your hands-on work revealed about how the technology actually works

Where do they agree? Where do they tension with each other? What does combining them reveal? These are the questions your Results section answers.

Results Structure

Lead with the Headline

The most important finding from your sources about AI in your domain

Key Applications

The most important applications your sources discussed

Exploration Observations

Specific things you observed during hands-on work

Confirmation

Where your exploration confirmed what the literature said

New Discoveries

Where your exploration revealed something the literature didn't address

A Quick Word on Honesty

Honest research includes what didn't work.

Don't shape your Results to make your hands-on exploration look more impressive than it was.

If the tool worked imperfectly...

Say so. Imperfect results are real results.

If something surprised you because it was worse than expected...

Say that. Negative surprises are often the most interesting findings.

Where the most interesting observations come from...

Often from what didn't work — that's where bias, safety, and ethical concerns hide.

Refining Your Related Work

After reading your sources deeply today, look back at the Related Work draft you wrote in S22 homework. You're not rewriting from scratch - you're sharpening based on what your deeper reading revealed.

More Specific Findings?

Are there findings you can now describe more specifically than you did in your draft?

Better Source Understanding?

Are there sources you understand better now that you should reference more accurately?

Still Synthesizing?

Is your synthesis still organized around patterns, or does it slip into source-by-source summary?

 Spend a few minutes refining. Small improvements here make a big difference in the final paper.

Production Block A

The first major block of today is the literature deep read.

Read Each Source in NotebookLM

Work through all 5–8 of your sources

Extract Specific Findings

Pull out examples, statistics, and concerns from each

Look for Patterns

Find where sources agree, disagree, and extend each other

Take Notes

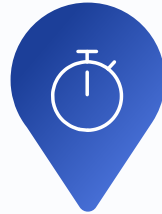
Record everything for Results and Related Work refinement

 Your mentor will check in at the midpoint and again at the end of this block.

Production Block B

Setup

Prepare tools and objectives



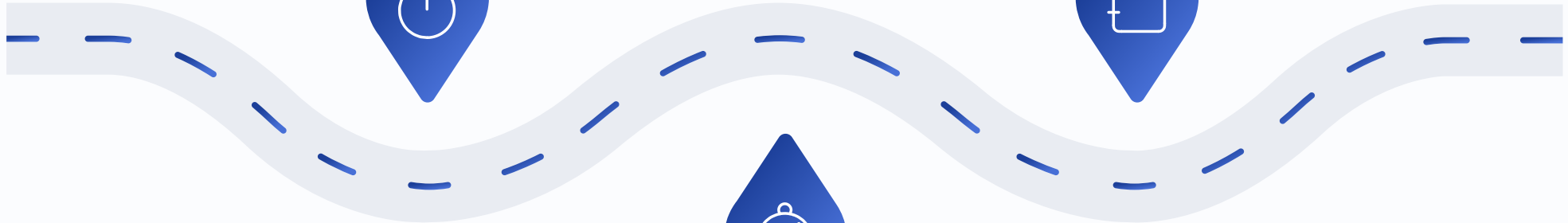
Document

Capture observations and notes



Exploration

Hands-on investigation with mentor



Production Block C

The final block is drafting Methodology and starting Results. Results doesn't have to be complete today - you'll expand it after your user testing in homework.

Draft your **Methodology section** - both literature and hands-on threads

Start your **Results section** with what you have so far - headline finding and key observations

 Results doesn't have to be complete today. You'll expand it after your user testing in homework.



User Testing Plan

For your homework, you'll do user testing - having **3-5 other people** interact with your hands-on exploration and giving you feedback.

Why It Matters

It shows how the technology behaves with people other than you. Their observations might surprise you and reveal things you missed.

Who Are Your Testers?

3-5 people, ideally a mix of ages or backgrounds

What Will They Try?

The same exploration you did, or a version of it

What Will You Ask?

Specific questions that get honest, useful responses

User Testing - What to Ask Your Testers

Useful questions to ask your testers. **Don't lead them to answers.** Let them tell you what they noticed.

“

"What did the tool seem to **do well**?"

”

“

"Where did it **confuse you**?"

”

“

"Did anything **surprise you**?"

”

“

"Would you **trust this** for [the domain's purpose]?"

”

“

"What would make you **trust it more**?"

”

Documenting User Testing

For each tester, write down the following. This documentation becomes part of your expanded Results section.



Who They Are

Age, role, any relevant context that might affect their perspective



What They Tried

Exactly what they did with the tool during the session



Their Reactions

In their own words where possible — don't paraphrase away the specifics



Patterns Across Testers

What multiple testers noticed - these are your strongest findings

Outputs Checklist

By the end of today's session, you should be able to check off every item below:

- ☐ Sources read deeply with specific findings extracted
- ☐ Hands-on exploration completed
- ☐ Observations documented
- ☐ Methodology section drafted
- ☐ Results section started
- ☐ Related Work refined
- ☐ User testing plan committed (3-5 testers, what they'll try, what you'll ask)

Homework Overview

Homework is focused on user testing and expanding Results.

Task
User testing with 3–5 testers; document observations from each tester
Expand Results section with testing observations (~150–250 words by grade)
Note 3 surprises or unexpected findings - these inform the Bias & Safety analysis next session

What's Next - Session 24

Bring to Session 24

- Your refined Related Work
- Your Methodology section
- Your expanded Results section
- Your user testing documentation
- Your three documented surprises

Session 24 - Analyze & Synthesize

You'll examine AI in your domain through **five lenses**, polish all your sections, and begin the final stretch:

- Draft Conclusion and final Abstract
- Write your AI Attribution Statement
- Start your presentation slides